

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 10: Altimeter Setting & Temperature Error

Syllabus Source: CAE Oxford (pp. 227-238) | Pass Target: >= 90% AviationExam

1. PRESSURE SETTINGS: QNH, QFE, STANDARD

- QNH (Altimeter setting): Station pressure reduced to MSL according to ISA. Indicates ALTITUDE above MSL. On the ground at the aerodrome, it indicates aerodrome elevation.
- QFE: Atmospheric pressure at aerodrome elevation (or runway threshold). Indicates HEIGHT above aerodrome datum. On the ground, it reads ZERO.
- Standard Setting (1013.25 hPa / 29.92 inHg): Used for vertical separation en-route. Indicates FLIGHT LEVEL (FL).

2. TRANSITION ALTITUDE, LEVEL & LAYER

Element	Authority / Publication	Operational Rule & Altimeter Reference
Transition Altitude (TA)	Published on instrument charts (Fixed)	Altitude at or below which vertical position is controlled by reference to ALTITUDES (QNH).
Transition Level (TRL)	Calculated and issued by ATC (Variable)	Lowest available FLIGHT LEVEL above the TA. Controlled by reference to Standard (1013.25). Varies with actual QNH.
Transition Layer	Airspace between TA and TRL	Airspace between TA and TRL (at least 1,000 ft thick). LEVEL CRUISING FLIGHT IN THIS LAYER IS STRICTLY PROHIBITED.

[!] EXAM TRAP (AVIATIONEXAM) - Altimeter Changeover Point

- In CLIMB: Altimeter is changed from QNH to STANDARD (1013.25 hPa) when PASSING THE TRANSITION ALTITUDE (TA).
- In DESCENT: Altimeter is changed from STANDARD (1013.25) to QNH when PASSING THE TRANSITION LEVEL (TRL).

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

Pressure altimeters are calibrated to the International Standard Atmosphere (ISA). In temperatures colder than ISA, the true altitude is LOWER than the indicated altitude ('High to low or hot to cold, look out below!').

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Altimeter Temperature Error and Terrain Clearance Correction

SCENARIO (PANS-OPS / EASA Cold Weather Altimetry Correction): - Aerodrome elevation: 2,000 ft. Published Minimum Safe Altitude (MSA): 6,000 ft. - Reported ambient temperature at aerodrome: -15°C. SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate Standard ISA Temperature at Aerodrome Elevation (2,000 ft): - ISA Temp = $15^{\circ}\text{C} - (2^{\circ}\text{C} \times 2) = +11^{\circ}\text{C}$. Step 2: Calculate ISA Deviation (Delta-ISA): - Delta-ISA = Actual (-15°C) - Standard (+11°C) = -26°C (Colder than ISA!). Step 3: Calculate Height Above Altimeter Source (Datum): - Height = 6,000 ft (MSA) - 2,000 ft (Elevation) = 4,000 ft (4 units of 1,000 ft). Step 4: Apply the 4 ft per 1,000 ft Rule of Thumb: - Error = $4 \text{ ft} \times 4 \times 26 = 416 \text{ ft}$. - Because air is colder than standard, True Altitude is 416 ft LOWER than indicated! FINAL ANSWER: To maintain the published 6,000 ft true clearance over obstacles, the pilot MUST fly an indicated altitude of at least 6,416 ft (6,000 + 416 ft)!

4. LOWEST USABLE FLIGHT LEVEL UNDER LOW PRESSURE

When atmospheric pressure drops below 1013.25 hPa, Flight Levels sink closer to terrain. For example, at QNH 980 hPa (33 hPa below standard): - $33 \text{ hPa} \times 27 \text{ ft/hPa} = \sim 891 \text{ ft}$ loss of true clearance. - ATC must raise the Transition Level and lowest usable Flight Level by 1,000 ft to preserve safety margins.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: In climb, when does the altimeter subscale setting change from QNH to Standard 1013.25?**

- [A]: At the Transition Level (TRL)
- [B]: Passing the Transition Altitude (TA)
- [C]: Leaving controlled airspace
- [D]: At 10,000 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. In climb, altimeters are set to 1013.25 upon passing the Transition Altitude (TA).

Q2: In air colder than ISA, does the altimeter overread or underread?

- [A]: It underreads (true altitude is higher than indicated).
- [B]: It overreads (true altitude is lower than indicated).
- [C]: It indicates correctly because pressure remains unchanged.
- [D]: It only errors when climbing above FL 100.

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. In cold air, air columns shrink; pressure levels drop closer to ground. The altimeter overreads (shows higher than actual true altitude).

Q3: Is level cruising flight permitted within the Transition Layer?

- [A]: Yes, if authorized by ATC.
- [B]: Yes, under VFR only.
- [C]: NO, level flight within the transition layer is strictly prohibited.
- [D]: Only during holding procedures.

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Cruising in level flight within the Transition Layer is strictly prohibited.

Q4: If QNH is 980 hPa (substantially below standard), what happens to the lowest usable Flight Level?

- [A]: It is lower than usual
- [B]: It is higher than usual to guarantee minimum obstacle clearance
- [C]: It is unaffected
- [D]: Flight Levels are suspended

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Low atmospheric pressure compresses pressure levels towards the surface, requiring ATC to raise the lowest usable Flight Level to maintain terrain clearance.

Q5: If an aircraft flies from an area of high pressure into an area of low pressure without adjusting altimeter setting, what happens?

- [A]: The altimeter will underread and aircraft will climb
- [B]: The altimeter will overread and true altitude will be dangerously lower than indicated
- [C]: Airspeed indicator will fail
- [D]: Autopilot will disconnect automatically

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. 'High to low, look out below!' The altimeter registers the lower ambient pressure as higher altitude, causing the pilot or autopilot to fly closer to terrain than indicated.

Q6: What is the maximum permitted tolerance when checking a primary altimeter against aerodrome elevation?

- [A]: +/- 20 ft
- [B]: +/- 60 ft (to +/- 75 ft depending on test conditions)
- [C]: +/- 150 ft
- [D]: +/- 250 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under PANS-OPS, the pre-flight ground altimeter check tolerance is +/- 60 to 75 ft of the published aerodrome/runway elevation.